



# INŽINIERI

## Rescue line

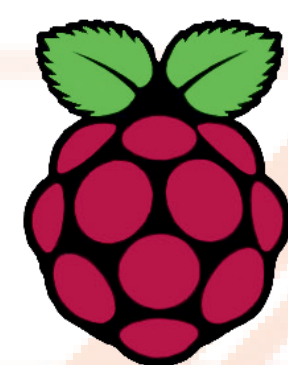


### Software

We use the Arduino IDE where we write code in C++ programming language. The software is divided into multiple modules based on their functionality, ensuring clear structure and easy maintenance. Custom functions, variables, and comments improve code efficiency and readability. A state-based control system manages the robot's behavior and enables switching between different tasks.

### Testing

All hardware components were tested under real competition conditions to verify their reliability and compatibility with the robot. Component testing was completed before the development of the main control logic.



RaspberryPi

### Robot Model

Our robot features a swing axle that helps it stay balanced on all four wheels with equal force. We also implemented a collection mechanism (gatherer) driven by a servo motor. On the gatherer are another servos that move with hands collecting the victims. Additional servos operate the mechanical hands that collect the victims, which the mechanism then transfers to the storage area located at the rear of the robot.

### 3D Design

The entire model of our robot was designed in Fusion 360, and development took us six months. The design of our robot is based on a previous version used in the latest RoboCup competition. We optimized the model of our robot for the best possible results and precise part fitting.

### About Us

We are a team of four students from a secondary vocational school of electrical engineering. We have been competing in the RoboCup for 5 years, winning the Slovak Championship in our category three times. We have also attended European Championship twice. Each member of our team has a specific role.

### Sebastián Janovčík

Responsible for programming the core program logic, calibration, sensor testing and problem-solving. He also developed the movement algorithms, intersection detections, and even the victim vision system.

### Andrej Gajdoš

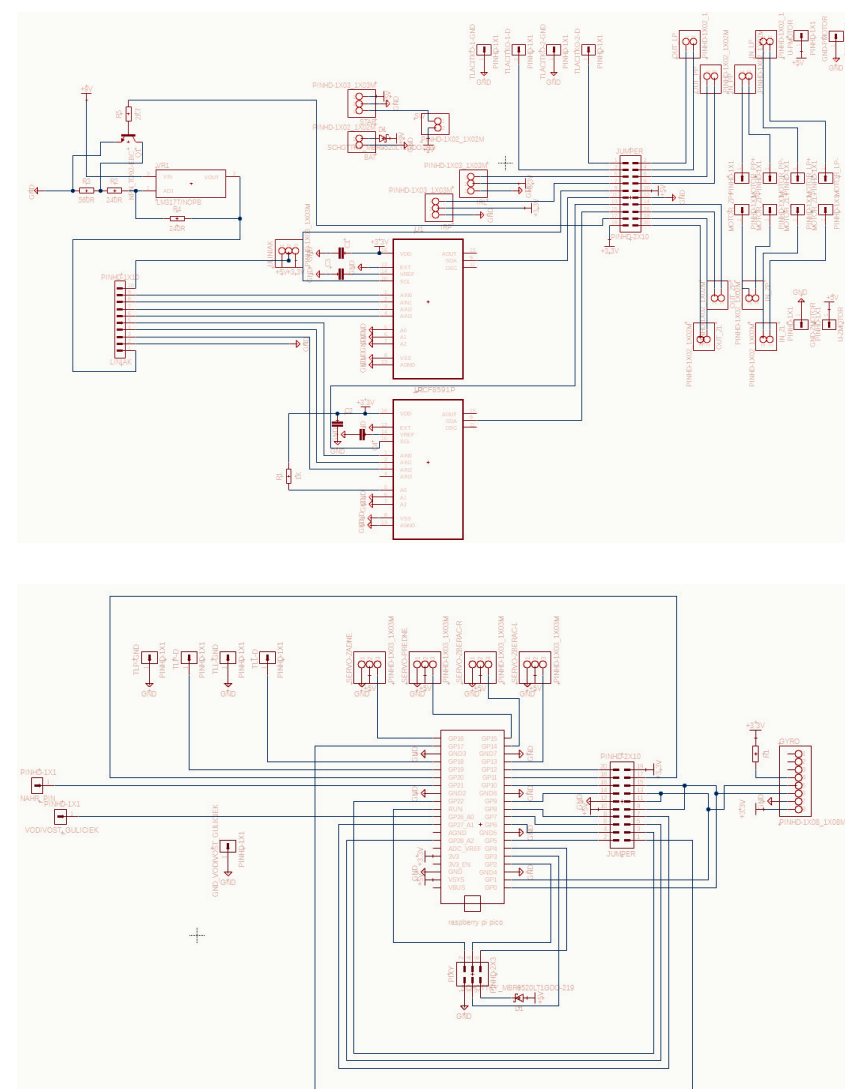
In charge of soldering electronic components, selecting suitable parts, and managing the design and development of PCB layouts and schematics. Oversees the integration and final assembly of the robot, ensuring proper functionality and reliability of all hardware systems. Also responsible for final system testing, debugging hardware issues, and preparing and delivering project presentations.

### Matúš Capko

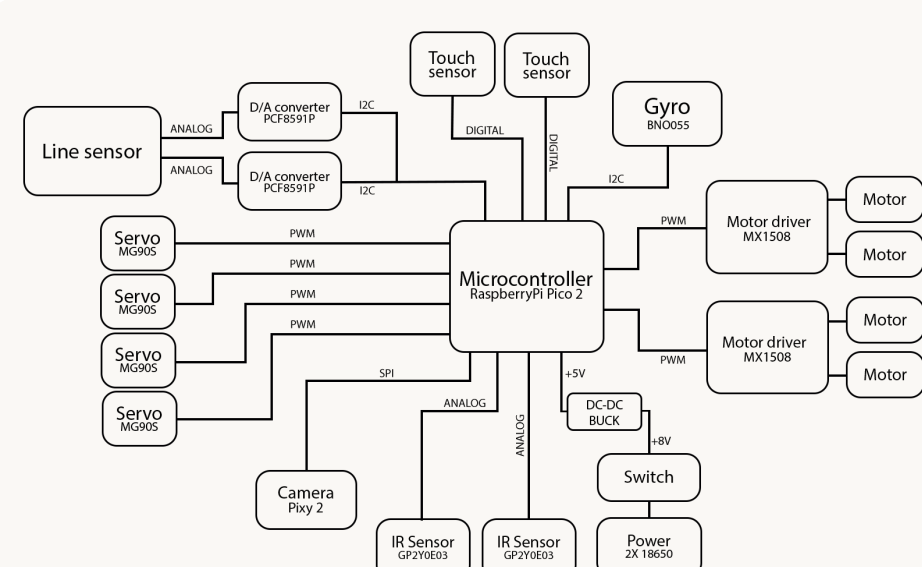
Lead designer of the robot's 3D models, who also created the schematics and PCBs. He was responsible for soldering and finalizing all hardware components, as well as developing the website.

### Sebastián Surovček

Responsible for developing the main program logic, calibrating the system, testing sensors, and finding solutions to technical problems, while also improving overall system performance and reliability.



| actual X-angle | Deg to turn | final x-angle |
|----------------|-------------|---------------|
| 350            | 90          | 80            |
| 0              | -40         | 320           |
| 230            | 180         | 50            |



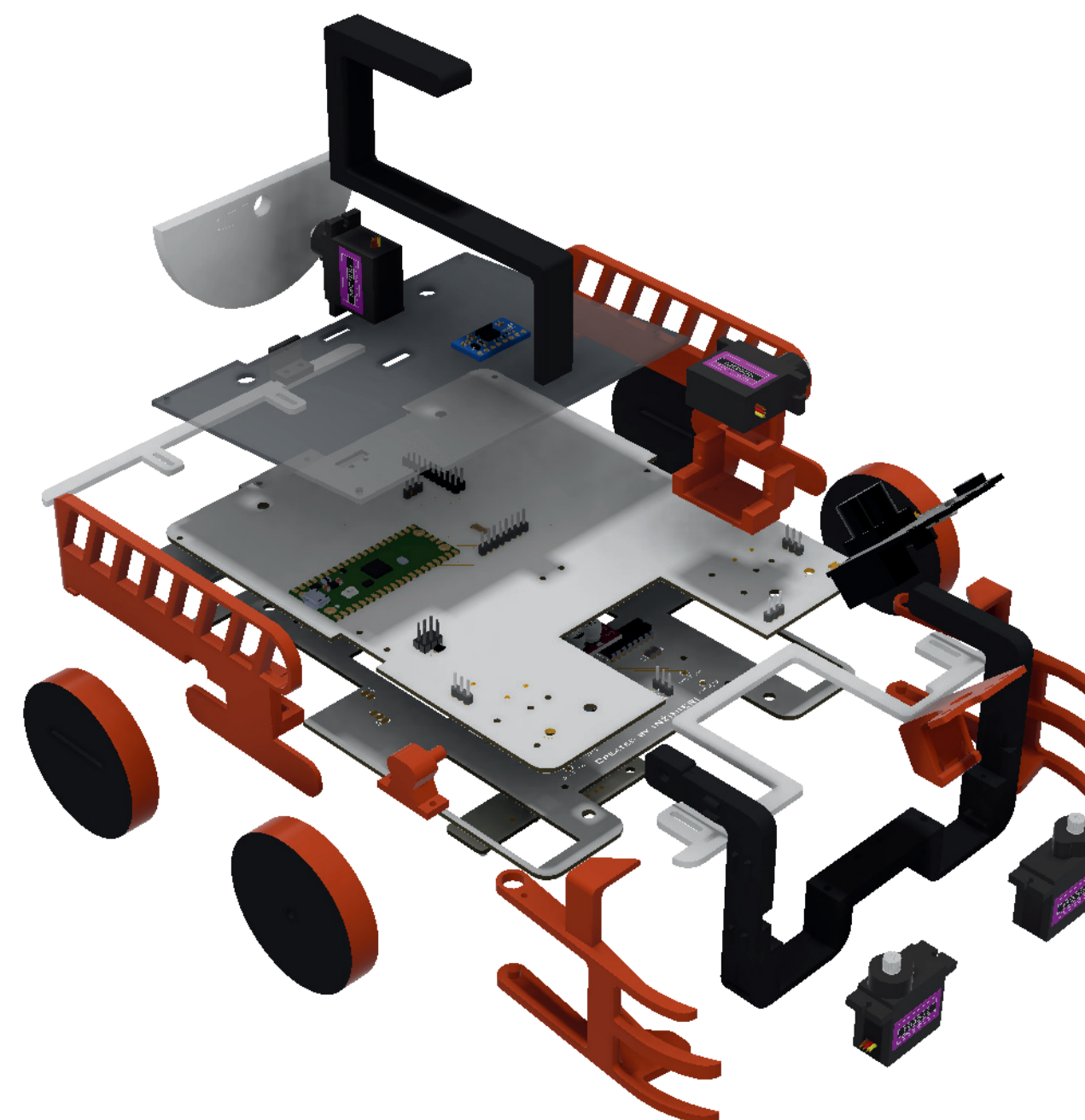
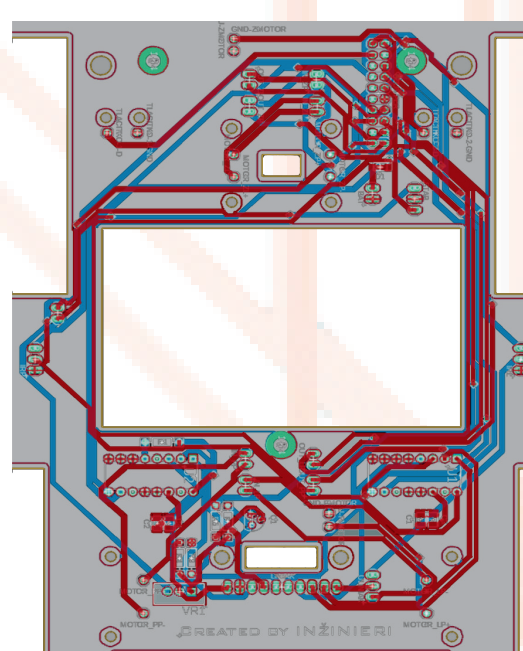
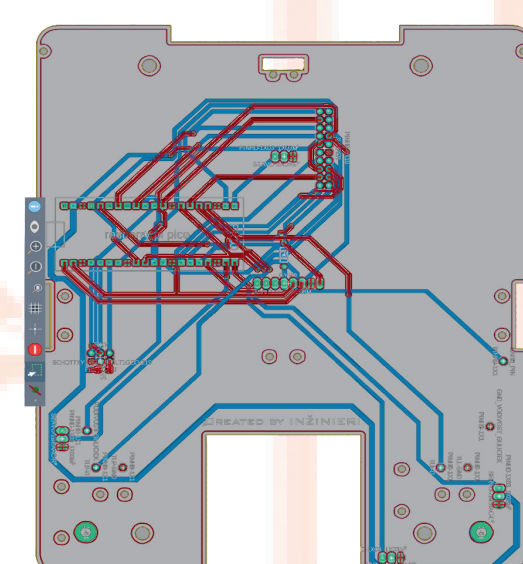
### Strategy

#### Line

The robot follows the line using a line sensor at the bottom of the robot, which contains 7 phototransistors. Intersections are evaluated by combining data from the line sensor and the camera. The gyroscope is helpful in certain situations, such as navigating a ramp or avoiding obstacles.

#### Evacuation Zone

Side-mounted infrared sensors provide reliable victim detection. The robot scans the evacuation zone while navigating around its perimeter. Detected victims are captured and collected. After collecting all victims, the robot sorts and places them into the correct evacuation points.



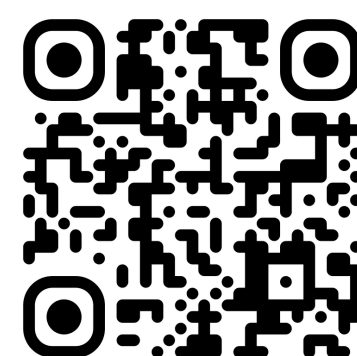
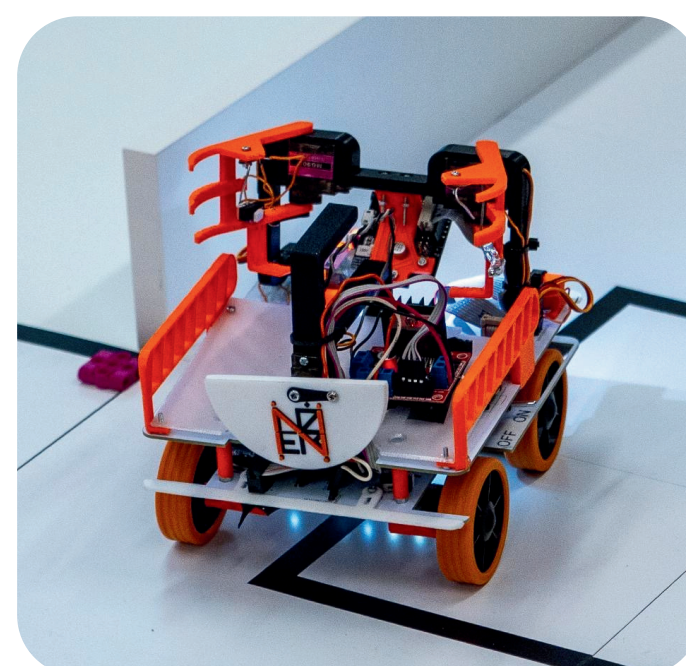
### Electrical Design

#### Electrical Parts

Raspberry pi pico 2 - for controlling the whole robot  
4x 6V 60rpm pololu motors - for robot movement  
4x mg90s servos - for gatherer movement  
2x MX1508 motor drivers - for motor control  
2x GP2Y0E03 IR sensors - for victims detection  
BNO055 orientation sensor - for orientation and better movement  
Pixy 2 camera - for green squares detection on intersections  
Self-made line sensor - for line following  
PCF8591 A/D converter module - for line sensor to run via I2C communication bus  
And others...

#### PCBs

We have decided to use PCB for eliminating wires and better design for our robot.



Our website



WEGA



DELTA

inoma comp

KOMUNITNÁ  
NADÁCIA  
LIPTOV

melodramatique

CORAgeo

GEO  
TRAVEL

DELTECH 35  
rokov na trhu

KOVO  
LIPTOV

urob  
si sám

VISSIM

AXENTA

IT-SPRAVA.sk

NÁRODNÉ  
CENTRUM  
ROBOTIKY

fibris

ŽILINSKÝ  
SAMOSPRÁVNÝ  
KRAJ

@ inzinieri\_sose